EXPERIMENT 06

Bigram N-Gram Text Generation

Aim

To implement a basic bigram model that predicts the next word from the current word and generates text.

Procedure

1. Prepare a small training sentence.
2. Split the sentence into words.
3. Build a dictionary of word-to-next-word transitions.
4. Start from a chosen word and follow the transitions.
5. Print the generated sequence.

Program

```
import random

text = "we learn python and we learn language models"
words = text.split()
bigrams = {}
for first, second in zip(words, words[1:]):
    bigrams.setdefault(first, []).append(second)

current = "we"
generated = [current]
for _ in range(8):
    choices = bigrams.get(current)
    if not choices:
        break
    current = random.choice(choices)
    generated.append(current)

print("Generated text:", " ".join(generated))
```

Output

```
OUTPUT — EXPERIMENT 06
Generated text: we learn python and we learn language models
(Note: output may vary because the next word is selected randomly.)
```

Result

Thus, a simple bigram model generated text by selecting words based on learned word transitions.